Curriculum Vitae

Personal Information

Ruisheng Cao
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Education

- Dr.Eng. in Computer Technology, Shanghai Jiao Tong University *2021-2025*
- Intern at XLANG Lab (with Prof. Tao Yu), the University of Hong Kong *Jan-Apr 2024*
- M.Eng. in Computer Technology, Shanghai Jiao Tong University *2018-2021*
- B.Eng. in Computer Science and Technology, Shanghai Jiao Tong University *2014-2018*

Honors and Awards

Wu Wenjun AI Honorary Doctoral Scholarship (15 PhDs in the same cohort) *2022-2025*
Outstanding Graduate of Shanghai Jiao Tong University *2021*
National Scholarship for Graduate Students *2018-2019, 2019-2020*
Zhiyuan Honor Degree of Bachelor of Science (48 undergraduates in SJTU) *2018*
Zhiyuan Honors Scholarship (top 5% students) *2014-2015, 2015-2016*
Academic Excellence Scholarship (First-Class, top 4) *2014-2015*

Projects

Qwen3-Coder and Qwen3-Coder-Next (2025/04-2026/01) Technical contributions include :

- **Pre-train :** implement best-fit-packing algorithm and its variants, and integrate them into the Megatron training framework seamlessly with C++ acceleration (40× faster than Python).
- **CPT and SFT :** Large-scale LLM- and verification-based data generation and environment synthesis for typical agentic coding tasks like SWE-bench and terminal-bench.
- **Tool call and chat templates :** Devise `qwen3_coder` XML-like Jinja template with the corresponding tool parser, and scale up the single format trajectory into 20+ tool chat templates.
- **Infrastructure support :** Key contributors of the rollout-evaluation pipeline across multiple agent scaffolds, and a suite of utility scripts for trajectory conversion and one-click deployment.

Evolvable Universal Digital Agent (2026/01-now) Current progress includes :

- **Evolution of BugMaker :** Starting from SWE tasks stemmed from single-atomic bugs, achieve controllable evolution at the data-level by composing multi-bugs within the same repository.
- `uda-mcp-server` **:** A package which leverages FastMCP library to enable one-command injection of customized tools and skills into arbitrary execution environments or sandboxes for agents.
- `qwen-agent-2.0` **:** Develop the next-generation Qwen-Agent framework with foundation built.

Selected Publications

See for full list.

- **Ruisheng Cao**, Mouxiang Chen, Jiawei Chen, Zeyu Cui, Yunlong Feng, Binyuan Hui, Yuheng Jing, Kaixin Li, Mingze Li, Junyang Lin, Zeyao Ma, Kashun Shum, Xuwu Wang, Jinxi Wei, Jiaxi Yang, Jiajun Zhang, Lei Zhang, Zongmeng Zhang, Wenting Zhao, and Fan Zhou (core contributors, listed alphabetically by last names). *Qwen3-Coder-Next Technical Report.* arXiv preprint arXiv :2603.00729, 2026.
- Mouxiang Chen, Lei Zhang, Yunlong Feng, Xuwu Wang, Wenting Zhao, **Ruisheng Cao**, Jiaxi Yang, Jiawei Chen, Mingze Li, Zeyao Ma, Hao Ge, Zongmeng Zhang, Zeyu Cui, Dayiheng Liu, Jingren Zhou, Jianling Sun, Junyang Lin, and Binyuan Hui. *SWE-Universe : Scale Real-World Verifiable Environments to Millions.* arXiv preprint arXiv :2602.02361, 2026.
- Lei Zhang, Mouxiang Chen, **Ruisheng Cao**, Jiawei Chen, Fan Zhou, Yiheng Xu, Jiaxi Yang, Liang Chen, Changwei Luo, Kai Zhang, Fan Yan, KaShun SHUM, Jiajun Zhang, Zeyu Cui, Hu feng, Junyang Lin, Binyuan Hui, and Min Yang. *MegaFlow : Large-Scale Distributed Orchestration System for the Agentic Era.* arXiv preprint arXiv :2601.07526, 2026.
- **Ruisheng Cao**, Hanchong Zhang, Tiancheng Huang, Zhangyi Kang, Yuxin Zhang, Liangtai Sun, Hanqi Li, Yuxun Miao, Shuai Fan, Lu Chen and Kai Yu. *NeuSym-RAG : Hybrid Neural Symbolic Retrieval with Multiview Structuring for PDF Question Answering.* Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1 : Long Papers), pages 6211-6239, Vienna, Austria, 2025.

Selected Publications

- Hongshen Xu, Zichen Zhu, Lei Pan, Zihan Wang, Su Zhu, Da Ma, **Ruisheng Cao**, Lu Chen and Kai Yu. *Reducing Tool Hallucination via Reliability Alignment.* Proceedings of the 42nd International Conference on Machine Learning (ICML), Vancouver, Canada, 2025.
- Fangyu Lei, Jixuan Chen, Yuxiao Ye, **Ruisheng Cao**, Dongchan Shin, Hongjin Su, Zhaoqing Suo, Hongcheng Gao, Wenjing Hu, Pengcheng Yin, Victor Zhong, Caiming Xiong, Ruoxi Sun, Qian Liu, Sida Wang, and Tao Yu. *Spider 2.0 : Can Language Models Resolve Real-World Enterprise Text-to-SQL Workflows ?* (**Oral, top 1.8% of submissions**) International Conference on Learning Representations (ICLR), 2025.
- **Ruisheng Cao**, Fangyu Lei, Haoyuan Wu, Jixuan Chen, Yeqiao Fu, Hongcheng Gao, Xinzhuang Xiong, Hanchong Zhang, Yuchen Mao, Wenjing Hu, Tianbao Xie, Hongshen Xu, Danyang Zhang, Sida Wang, Ruoxi Sun, Pengcheng Yin, Caiming Xiong, Ansong Ni, Qian Liu, Victor Zhong, Lu Chen, Kai Yu and Tao Yu. *Spider2-V : How Far Are Multimodal Agents From Automating Data Science and Engineering Workflows ?* (**Spotlight**) Advances in Neural Information Processing Systems (NeurIPS), 2024.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, **Ruisheng Cao**, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong and Tao Yu. *OSWorld : Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments.* Advances in Neural Information Processing Systems (NeurIPS), 2024.
- Hanchong Zhang, **Ruisheng Cao**, Hongshen Xu, Lu Chen and Kai Yu. *CoE-SQL : In-Context Learning for Multi-Turn Text-to-SQL with Chain-of-Editions.* In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics : Human Language Technologies (NAACL), pages 6487-6508, Mexico City, Mexico.
- **Ruisheng Cao**, Lu Chen, Jieyu Li, Hanchong Zhang, Hongshen Xu, Wangyou Zhang and Kai Yu. *A Heterogeneous Graph to Abstract Syntax Tree Framework for Text-to-SQL.* In IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023 (01) : 1-16.
- **Ruisheng Cao**, Lu Chen, Zhi Chen, Yanbin Zhao, Su Zhu and Kai Yu. *LGESQL : Line Graph Enhanced Text-to-SQL Model with Mixed Local and Non-Local Relations* (**Long Oral**). In 59th Annual Meeting of the Association for Computational Linguistics (ACL) and the 11th International Joint Conference on Natural Language Processing, pages 2541-2555, 2021.
- Zhi Chen, Lu Chen, Yanbin Zhao, **Ruisheng Cao**, Zihan Xu, Su Zhu and Kai Yu. *ShadowGNN : Graph Projection Neural Network for Text-to-SQL Parser.* Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics : Human Language Technologies (NAACL), online, pages 5567-5577, 2021.
- **Ruisheng Cao**, Su Zhu, Chenyu Yang, Chen Liu, Rao Ma, Yanbin Zhao, Lu Chen and Kai Yu. *Unsupervised Dual Paraphrasing for Two-stage Semantic Parsing* (**Long Online Oral**). In 58th Annual Meeting of the Association for Computational Linguistics (ACL), Seattle, USA, pages 6806-6817, 2020.
- **Ruisheng Cao*** (equal contribution, responsible for code), Su Zhu* and Kai Yu. *Dual Learning for Semi-Supervised Natural Language Understanding.* IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 28, pp. 1936-1947, 2020.
- Yanbin Zhao, Lu Chen, Zhi Chen, **Ruisheng Cao**, Su Zhu and Kai Yu. *Line Graph Enhanced AMR-to-Text Generation with Mix-Order Graph Attention Networks* (**Long Online Oral**). In 58th Annual Meeting of the Association for Computational Linguistics (ACL), Seattle, USA, pages 732-741, 2020.
- **Ruisheng Cao**, Su Zhu, Chen Liu, Jieyu Li and Kai Yu. *Semantic Parsing with Dual Learning* (**Long Oral**). In 57th Annual Meeting of the Association for Computational Linguistics (ACL), Florence, Italy, pages 51-64, 2019.

Challenges And Competitions

Rank 1	DuSQL : A Large-Scale and Pragmatic Chinese Text-to-SQL Dataset.	*2021/10-present*
Rank 1	Spider : Yale Semantic Parsing and Text-to-SQL Challenge.	*2021/06-2022/09*
Rank 1	CSpider : The Chinese Semantic Parsing and Text-to-SQL Challenge.	*2022/05-2023/06*
Rank 1	NL2SQL : A Large-Scale Chinese Single-Table Text-to-SQL Dataset.	*2022/04-present*